

Initial report on Golap Solar PV plant

Prepared for Zilla Parishad, Ratnagiri

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IIT Bombay, 20 February, 2025

This report contains::

- A. Some preliminary assessment of the functioning of the plant
- B. Guidelines for quality and reliability.

A. Preliminary analysis of current performance

This analysis is based on data from the SCADA system. If there are any issues this will show up in the data. MSEDCL metered energy data will be an important cross check to conduct.

1. Current performance as compared to the DPR

1.1 DPR configurations and generation projections

The plant has been operational since August 2024. Table 1 shows the generated energy data for four months. from the SCADA system, projected generation from the DPR in year 10 of the plant, and a simulation for the installed configuration conducted on PVSyst

Table 1. Generated energy, DPR projections, and simulations. The energy numbers are in kWh

A	B	C	D	E
Month	Generated (2024)	DPR (1.2 MW) Yr 10	DPR (1.0 MW) Yr 10	Installed configuration Yr 10
Sep	97231	146380	148230	115915
Oct	104924	140663	156548	123332
Nov	114982	179778	146398	133099
Dec	111415	183835	146399	118217

Column B has the generated energy. Column C and D show figures from the DPR. There were two sets of figures in the DPR . Column C is for a 1.2 MW plant, that may have been an initial

design. Column D is for a 1 MW plant - it is not clear what input configuration was given to PVSyst for the 1 MW projection. The 1.2 MW simulation done in PVSyst (column C) has a different set of panels and inverter (Jinko Solar panels and Sungrow inverter) from what is finally installed.

Column E has the 10 year simulation for the installed configuration conducted by IIT: Bluebird panels and Huawei inverters. This simulation is very different from the 1 MW projection in the DPR. Also, the 1 MW and 1.2 MW projections in the DPR are not very different. The DPR should reflect the installed configuration.

1.2 PVSyst Simulations for installed configuration

Table 2 shows the generated energy data from the SCADA system, and simulations for various configurations.

Column F has the simulation for year 1 for the installed system. The comparison of year 1 expected output and actual generated output is in column G. It is clear that the generation is much lower than the simulated for November and December. For September and October there was no data for several days on the SCADA system. Column H shows the approximate hours with zero generation. It is not clear what this is due to. We must also note that if the data from the SCADA system has some issues, then the generated energy displayed will not be accurate. It is better to do a comparison with data as recorded by MSEDCL.

The difference in generation and simulation may be due to tracking issues and loss of grid availability during hours of generation. Some high level preliminary analysis is conducted below.

Table 2. Generated energy, simulations for year 1. The energy numbers are in kWh

A	B	F	G	H	I	J
Month	SCADA data (2024)	Installed configuration	Installed Generated less than the simulation	Zero generation hrs (approx. number)	Fixed tilt	Installed, Generated less than fixed tilt simulation
Sep	97231	1,22,770	26.27%	19%	91,843	-5.54%
Oct	104924	1,30,622	24.49%	23%	1,16,062	10.62%
Nov	114982	1,40,962	22.59%	2%	1,43,406	24.72%
Dec	111415	1,25,203	12.38%	1%	1,46,901	31.85%

Total for a year		16,58,533			15,33,412	
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2. Tracking

The system is set up at an axis title of 0° with the tracking from East to West limited to $\phi = \pm 15^\circ$. (This means that the normal to the panel changes angle over a sector of 30° , $\pm 15^\circ$ to the normal to a horizontal plane). See figure 1.

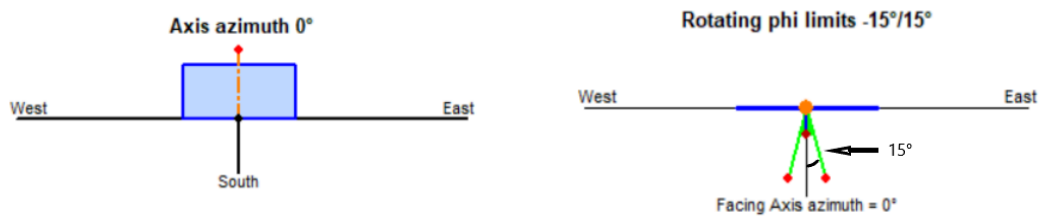


Figure 1. Axis tilt in installed panels

Observations showed that the tracking mechanisms were not functioning properly - see figure 2. With the various tables at different angles. The loss due to tracking issues can be high and are the likely reason for the low energy generated.



Figure 2. Pictures taken at the plant on January 8th 2025

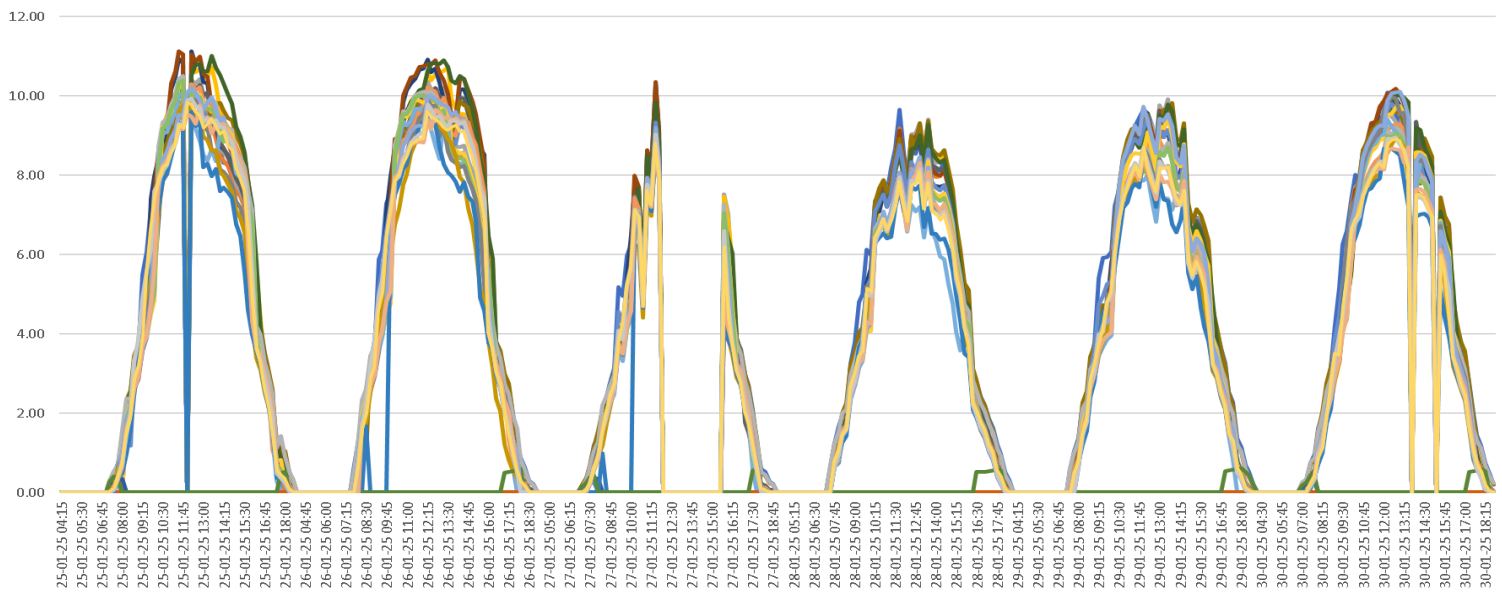


Figure 3. Generated current from all strings of inverter 1 from January 25th to 30th, 2025

The current data taken for a few sample days in figure 3 shows that some strings are consistently underperforming (yellow) and some are high (green, brown). An analysis of the average currents by string for inverter 1 from Jan 25 to Feb 8 showed that strings 1, 3, 4, 10 are constantly in the top generating 5 to 6 strings and strings 23, 26, 28 are mostly in the bottom 5. There is a difference of 10 to 20% in the top and bottom generators (Analysis in Appendix A). This possibly indicates a tracking issue with certain tables.

The above analysis and attribution to tracker malfunction is only an estimate since SCADA data could be tainted by several factors. However, it has been reported with increasing frequency from the field that the tracker-misalignment can cause significant reduction in the yield.

Since these installations are quite near the ocean and in hot and humid areas, we would like to recommend that you do not use trackers (which can be highly prone to corrosion and in case of Golap plant also mis-alignment) but go with fixed latitude tilt systems.

A simulation of the current configuration and placement of panels, with Fixed tilt panels with no tracking, at an optimum angle of 23 is listed in column H of table 1. The simulation for the fixed tilt panels shows higher than actual generation for three out of four months.

Also, the simulated generation from a fixed tilt system (1533 MWh) is 8% less than the simulated generation of the installed system (1659 MWh) over a year. The cost of the unipolar tracking structure, as per the DPR is 55 lakhs (DPR for 1 MW plant), over a total cost of ~7 crores (EC DPR GOLAP), which is 8 % of the total cost. Financially, in the current installation of the trackers, even if they were working perfectly, may not be adding value. It may also be noted that with misalignment, the trackers may leave the panels in a configuration worse off than the fixed tilt especially if there is no axis tilt as in this case.

B. Guidelines for Quality and Reliability

I. Data Requirements from Module Manufacturer / Developer:

1. Data sheet of all types of modules to be used in the plant should be shared in advance. This shouldn't be a 'representative module' but the exact module to be used by the developer.
2. They should also share a PAN file of the exact module which would be used to run PVSyst or similar simulations.
3. All the assumptions made in the PVSyst simulations should be explicitly stated and these should be as realistic as possible. For example, the exact inverter make and model, the exact tracking algorithm, soiling rates, cleaning frequency etc. used during simulations in PVSyst should be explicitly shared.
4. All the necessary certifications passed by the module should also be shared. Actual certificates from the test labs should be made available.
5. Detailed Bills of Materials - particularly including the exact composition of encapsulant and backsheet / back glass should be shared. If they are using a glass - glass module package, then they should clearly specify the glass thickness and if the front / back glass are tempered or just heat strengthened. The datasheets of glass, encapsulant, cells, junction box and backsheet should be shared.
6. The exact I-V data (preferably entire I-V curve, but if it's not available, then the actual measured I-V data) measured at the factory for each module should be shared before the modules are shipped. Ideally this should include the I-V curve, but at minimum, it should include the Pmax, Isc, Voc, Fill Factor. This information is critical to verify if the plant performance is subpar at some point of time and if the modules are eligible for warranty returns. This data is available with manufacturers.

7. The manufacturer captures the Electroluminescence image of every module produced on their line before shipment. These images can help in tracing the defects present in the modules and identify if those were present from the manufacturing facility or were newly generated during the installation / operation and maintenance. This data already exists with the manufacturers and it should be shared.
8. The developer should also share the details of locations and customers where the chosen modules have been deployed in the field in India. Total amount of field experience of these modules in India is important and customers who are already using these modules should be consulted before placing an order for a large quantity of modules.
9. If the modules have RFID tags, then appropriate RFID readers should be made available at the plant.

II. Recommended Module Certifications:

1. Note that IEC stands for International Electrotechnical Commission and IS stands for Indian Standard. There are equivalent IS standards for the corresponding IEC standards. Either of the standards is fine for a given purpose.
2. It is mandatory that all modules sold in India are certified with the design qualification standard IEC 61215 (IS 14286) and safety standard IEC 61730 (IS 61730). Since this requirement has been there for a while, it's unlikely that you would find a module that doesn't satisfy these criteria.
3. Modules that are deployed near the ocean are likely to experience salt spray and therefore, in such situations, one should insist that they are certified with the Salt Mist Corrosion Test (IEC / IS 61701). This would ensure a minimum level of durability in the modules to withstand salt spray from the ocean. This is not mandatory and you would have to request a module that's passed this test.
4. Modules that are deployed in agricultural fields with a lot of cattle and other livestock present, should be certified with the Ammonia Corrosion Test (IEC 62716 / IS 16664). This would ensure a minimum level of durability against corrosion caused by Ammonia which can be generated via excreta of the animals. This is not mandatory and you would have to request a module that's passed this test.
5. Modules should also be certified against the Potential Induced Degradation (PID) as per IEC TS 62804. PID has been shown to be one of the most commonly observed

degradation modes in the modules deployed in India. This standard is still optional but it is strongly recommended that the modules are certified as per this standard.

III. Preliminary Analysis of SCADA Data

1. Preliminary analysis of the SCADA data and images from the plant reveal that tracker misalignment is a big issue that's causing mismatch in the strings and this is also the likely cause behind discrepancy in the predicted generation (using PVSyst) and the actual generation from a PV plant.
2. We have also observed that there were issues in the timestamps in the SCADA data which were resulting in logged generation during night - time.
3. Having a weather station at the site is a must if we want to assess Performance Ratio (PR), Soiling rates etc. Currently, due to the lack of a weather station, PR estimation is not possible, and we will have to rely on some non-accurate, representative satellite data for that location.

IV. Recommended Requirements at the Site

1. The site should be equipped with a weather station and should have following measurement capabilities: Plane of Array Irradiance, Ambient Temperature, Module Temperature, Relative Humidity.
2. The weather data should be recorded every minute (to ensure a match with the SCADA data and also to record variability).
3. Preferably, the SCADA generation data should be monitored at the string level and should be recorded at a frequency of at least every 10 minutes.
4. It has been reported with increasing frequency from the field that the tracker-misalignment can cause significant reduction in the yield. It's quite difficult to detect this from SCADA data as it could be tainted by several factors. Therefore, having a high resolution, remote-controllable visual camera system that could see the entire plant is highly recommended. This way, one could remotely detect potentially mis-aligned strings and these could be corrected. Otherwise, if it is possible from the trackers, data on the alignment of the panels should be logged.
5. Since these installations are quite near the ocean and in hot and humid areas, we would like to recommend that you don't use trackers (which can be highly prone to corrosion and in case of Golap plant also mis-alignment) but go with fixed latitude tilt systems.

Appendix A

Table A.1 Sum of currents (15 minute ntervals) for each string over 6 days. Averages have been taken only for Jan 31 and Feb 1

Sum of currents (15 minute intervals) for each string over 6 days												Averages taken for Jan 31, Feb 1															
Top 5 strings average	286		286		248		248		229		123		4.45		5.39		267		275		267		269		281		226
String	25 Jan	String	26 Jan	String	27 Jan	String	28 Jan	String	29 Jan	String	30 Jan		Jan 31		Feb 1		3 Feb		4 Feb		5 Feb		6 Feb		7 Feb		8 Feb
1	302	1	301	1	260	1	259	1	238	1	129	1	5	1	5	1	278	1	289	1	276	1	280	1	296	1	237
4	285	4	286	4	247	10	250	10	234	10	125	10	4	2	5	4	266	4	276	3	268	4	268	10	282	4	226
3	285	3	285	19	245	19	247	19	231	19	124	19	4	3	5	3	264	3	271	10	267	3	268	4	280	10	226
10	281	10	281	10	245	4	245	4	223	4	118	15	4	4	5	19	264	10	270	4	264	10	267	3	280	3	224
19	276	19	277	15	244	3	242	7	221	3	117	4	4	7	5	15	264	15	268	15	260	19	261	15	268	8	216
7	274	7	273	3	244	7	241	8	220	7	116	3	4	8	5	10	263	19	267	19	259	15	259	19	266	19	215
17	272	2	269	7	237	8	238	3	220	15	114	7	4	10	5	7	254	17	262	7	253	8	253	2	265	2	215
16	271	8	268	2	234	17	233	17	216	22	113	8	4	15	5	8	252	7	262	9	252	2	252	8	261	15	214
2	270	17	264	8	233	22	232	14	216	17	112	22	4	9	5	2	250	2	259	14	252	14	251	7	260	7	214
8	267	16	262	18	230	14	232	22	216	14	111	17	4	14	5	17	249	8	259	12	250	7	251	14	259	12	212
27	264	22	262	17	230	2	231	9	215	9	111	18	4	13	5	28	247	18	258	8	250	12	250	12	258	14	211
22	262	27	261	22	229	9	231	16	213	8	111	2	4	12	5	18	246	28	255	2	248	9	249	9	257	9	211
15	262	9	261	14	227	16	230	12	212	2	111	9	4	16	5	22	245	16	255	17	247	17	247	22	256	17	208
9	260	14	260	16	227	12	229	2	212	12	110	14	4	17	5	16	245	26	255	18	246	16	243	17	254	22	208
14	260	12	257	9	226	27	228	21	211	16	110	16	4	18	5	14	245	22	254	16	246	22	243	18	252	16	206
12	256	21	255	12	225	13	227	27	210	21	109	12	4	19	5	9	244	27	253	22	245	18	242	16	251	27	204
21	256	15	249	27	224	21	227	15	210	27	108	27	4	22	5	12	243	12	253	26	244	26	239	26	248	18	204
18	248	13	245	13	223	15	223	13	203	13	105	13	4	21	5	26	243	14	252	27	243	27	238	27	247	21	203

13	245	28	243	21	222	18	214	18	199	18	102	21	4	23	5	27	243	9	251	28	243	28	238	28	246	26	202
28	243	18	241	28	210	28	212	28	196	28	102	28	4	26	5	23	242	21	250	13	238	21	237	21	245	28	201
26	239	26	238	26	206	23	208	23	196	23	100	23	4	27	5	21	239	13	248	21	237	23	237	13	245	13	199
23	236	23	236	23	205	26	207	26	193	26	100	26	4	28	5	13	239	23	247	23	236	13	236	23	243	23	197
Last 5 strgs Avg	242		240		215		214		197		102		3.88		5.15		241		250		240		238		246		201
Diff betwn top 5, last 5	18%		19%		15%		16%		16%		21%		15%		5%		11%		10%		11%		13%		14%		13%

Table A.2 Indicates the consistency of a string in the top or bottom.

Table A.2 Frequency of a string in the top 5 or bottom 5 over 15 days

Top			Bottom	
SString no.	Occurrence		String no.	Occurrence
1	15		21	9
4	15		13	13
3	12		28	13
10	13		26	12
19	11		23	15